

Advertisement Creator – Using Image Generation (HPPCS[03])

Abstract

This project develops an automated advertisement-creation pipeline that converts three advertisement text inputs and their supplied advertisement templates into three final advertisements with realistic product photography. Qwen3 is used as LLM #1 to create a structured creative brief, while Llama 3.2 is used as LLM #2 to convert that brief into a FLUX-ready image prompt. FLUX.1-schnell generates the realistic product/lifestyle visual, and Python/Pillow inserts that visual into the supplied template while preserving the exact advertising copy. This separation avoids unreliable AI-generated typography and produces a professional advertisement rather than a randomly drawn graphic.

1. Introduction

Generative AI can reduce the manual effort required to transform marketing copy into professional advertising visuals. This project demonstrates an end-to-end local/open-model workflow in which two LLMs perform different reasoning tasks, an image-generation model produces realistic commercial photography, and deterministic Python post-processing assembles the final advertisement.

2. Problem Statement

The task is to transform three supplied advertisement concepts into three polished advertisements. The solution must interpret the marketing message, create a suitable visual concept, generate realistic product imagery, and place that imagery into the supplied layouts without losing exact headline, message and CTA text.

3. Objectives

- Use two LLMs with clearly different responsibilities.
- Generate realistic commercial product/lifestyle imagery instead of random placeholder graphics.
- Use the supplied advertisement layouts as the final composition templates.
- Preserve exact advertisement copy through deterministic Pillow composition.
- Produce three final advertisement PNG files.
- Provide both a demo mode and a full local AI generation mode.

4. Methodology

The pipeline reads the three advertisement text files and reuses the single supplied advertisement template for each output. Qwen3 analyzes the text and creates a creative brief covering product, audience, objective, tone, setting, lighting and composition. Llama 3.2 converts the brief into a detailed image prompt and

explicitly instructs the image model not to create copy or typography. FLUX.1-schnell generates only the realistic product/lifestyle visual. Pillow then replaces the template's visual placeholder with the generated image. The supplied template copy remains deterministic and exact.

5. System Design / Implementation

The Codebase is intentionally flat. main.py contains the complete orchestration. qwen3:8b is the first LLM, llama3.2:3b is the second LLM, and FLUX.1-schnell is the image model. ad_template.png is the single supplied advertisement template used for all three outputs. The template is based on the supplied actual-ad visual style and is reused consistently for the three advertisement texts. ad1_generated_visual.png, ad2_generated_visual.png and ad3_generated_visual.png are realistic generated sample visuals. final_ad_1.png, final_ad_2.png and final_ad_3.png are the final composed advertisements. requirements.txt, execution.txt, sample_prompts.json and generation_log.json support reproducibility and evaluation.

6. Results and Analysis

The revised implementation produces visually realistic advertisements rather than flat/random drawings and uses one actual advertisement template consistently for all three outputs. The GlowBrew advertisement uses a premium cold-brew bottle and iced coffee scene; PureNest uses an eco-friendly laundry-sheet package with towels and natural surroundings; AeroFit uses a premium smartwatch in a mountain fitness setting. These visuals are inserted into the supplied templates while the exact headline, supporting message and CTA remain controlled by Pillow. This provides a clear separation between creative image generation and deterministic advertising layout.

7. Conclusion and Future Work

The solution demonstrates a practical multi-model advertisement-generation workflow using Qwen3, Llama 3.2, FLUX.1-schnell and Pillow. Future improvements can include automatic visual-quality scoring, brand-guideline enforcement, multiple ad sizes, image inpainting/editing, background removal, product-reference conditioning, A/B variants and comparison of FLUX.1-schnell with SDXL or HiDream-I1.

9. References

Hugging Face Diffusers Documentation – <https://huggingface.co/docs/diffusers/>

Black Forest Labs – FLUX.1-schnell – <https://huggingface.co/black-forest-labs/FLUX.1-schnell>

Qwen – <https://qwenlm.github.io/>

Meta Llama – <https://www.llama.com/>

Ollama – <https://ollama.com/>

Acknowledgement

I acknowledge the HAAI++ capstone project guidelines and the supplied project materials that defined the advertisement inputs, template-driven workflow, two-LLM requirement and submission structure.